Document Title: Real-Time Myanmar Sign Language (MSL) to Text Inference Engine

Project Name: Warm Hands — *Hands That Connect Hearts*

Team Name: Fantastic 3 (Myanmar Institute of Information Technology - MIIT)

Target Event: AI Ready ASEAN Youth Challenge 2026

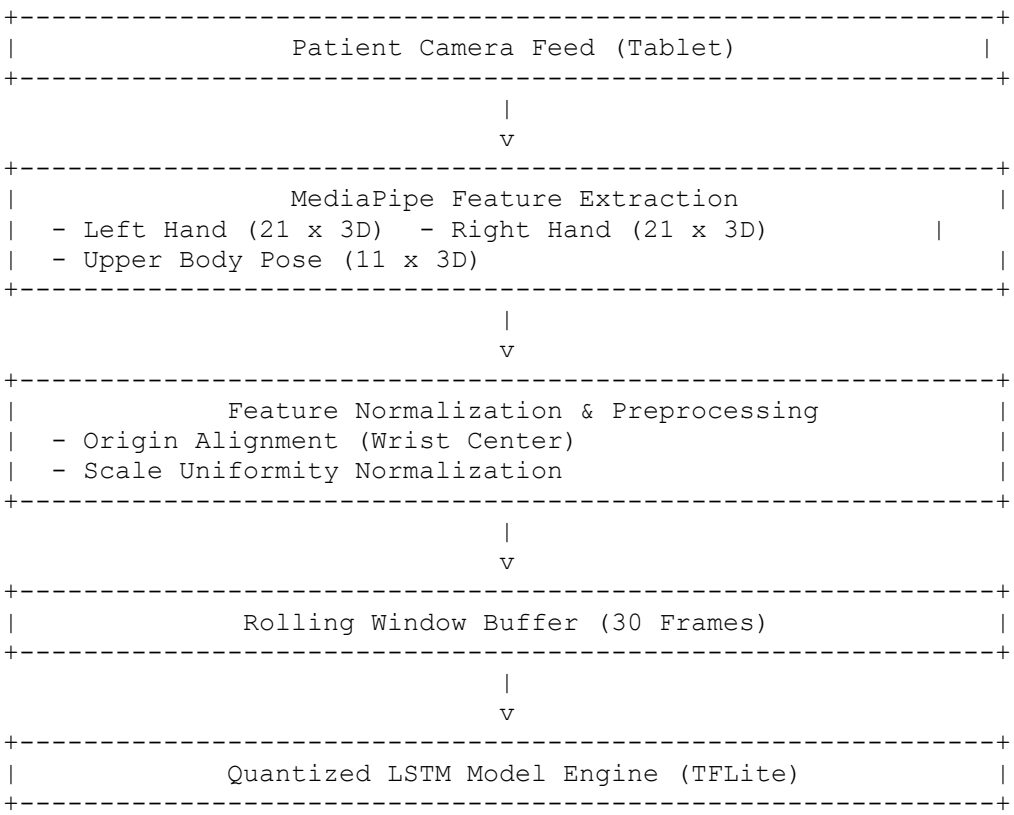
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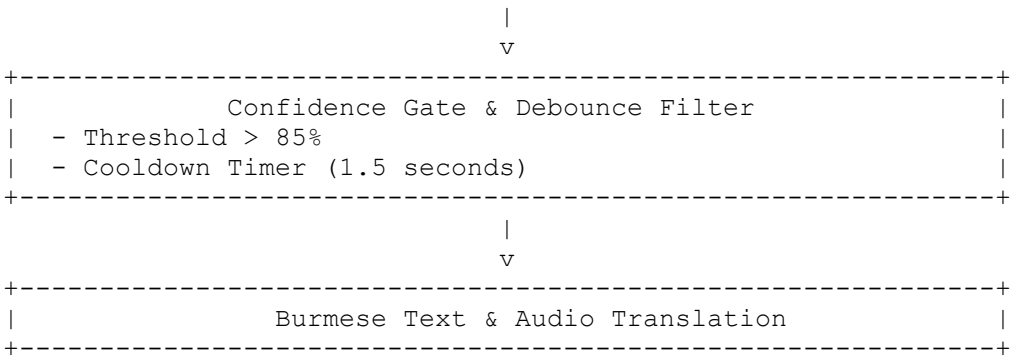
Date: August 2026

1. Executive Summary

The **Warm Hands** project is designed to bridge the severe communication gap between healthcare providers and deaf, hard-of-hearing, or speech-impaired patients in Myanmar. Due to the absence of publicly available Myanmar Sign Language (MSL) datasets, this subsystem establishes a complete data collection protocol, keypoint landmark extraction via **Google MediaPipe**, dynamic temporal pattern recognition using **LSTM Neural Networks**, and low-latency edge deployment using **TensorFlow Lite**.

2. System Workflow & Data Pipeline Architecture





3. Phase 1: Vocabulary Design & Data Collection Protocol

3.1 Medical Sign Vocabulary (100 Core Gestures)

Signs are selected in collaboration with MSL experts and categorized into key clinical intake areas:

1. **Symptoms & Sensations:** Fever, Dizziness, Sharp Pain, Dull Ache, Nausea, Breathing Difficulty, Swelling, Numbness.
2. **Anatomical Locations:** Head, Chest, Abdomen, Back, Throat, Joint, Left Arm, Right Leg.
3. **Scale & Timeline:** Pain Level (Scale 1–10), Morning, Night, Sudden, Continuous.
4. **Emergency Statements:** Medical History, Allergy, Need Assistance, Heart Problem.

3.2 Data Collection Protocol Specifications

Parameter	Specification Criteria	Technical Rationale
Participant Count	15–20 distinct signers	Prevents overfitting to specific hand anatomies or signing styles.
Camera Distances	0.5 meters to 1.5 meters	Covers mobile, desktop, and bedside tablet positioning.
Lighting Setup	150 Lux to 800 Lux	Assures landmark tracking integrity under varied hospital lighting conditions.
Captures per Class	30 sequence video clips	Yields 3,000 baseline clips across 100 classes.

Parameter	Specification Criteria	Technical Rationale
Video Format	30 FPS at 720p (1280x720)	Standardizes frame sequence sampling.

4. Phase 2: Landmark Feature Extraction

Instead of feeding high-dimensional raw video frames into the machine learning model, positional keypoints are extracted frame-by-frame using **Google MediaPipe**.

4.1 Frame Feature Vector Topology

- **Left Hand Keypoints:** 21 landmarks $\times 3\text{ coordinates } (x, y, z) = 63\text{ features}$.
- **Right Hand Keypoints:** 21 landmarks $\times 3\text{ coordinates } (x, y, z) = 63\text{ features}$.
- **Pose / Upper Body Keypoints:** 11 landmarks $\times 3\text{ coordinates } (x, y, z) = 33\text{ features}$.
- **Total Feature Vector Dimension per Frame:** $63 + 63 + 33 = 159\text{ numerical values}$.

4.2 Landmark Extraction Script (Python)

Python

```
import cv2
import numpy as np
import mediapipe as mp

mp_holistic = mp.solutions.holistic

def extract_landmarks(results):
    """
    Extracts and flattens left hand, right hand, and upper pose keypoints.
    Fills zero-arrays if a landmark set is not detected in the frame.
    """
    # Left Hand
    if results.left_hand_landmarks:
        lh = np.array([[res.x, res.y, res.z] for res in
results.left_hand_landmarks.landmark]).flatten()
    else:
        lh = np.zeros(21 * 3)

    # Right Hand
    if results.right_hand_landmarks:
        rh = np.array([[res.x, res.y, res.z] for res in
results.right_hand_landmarks.landmark]).flatten()
```

```
    else:
        rh = np.zeros(21 * 3)

    # Upper Pose (Landmarks 0 to 10: Head, Shoulders, Arms)
    if results.pose_landmarks:
        pose = np.array([[res.x, res.y, res.z] for res in
results.pose_landmarks.landmark[:11]]).flatten()
    else:
        pose = np.zeros(11 * 3)

    return np.concatenate([lh, rh, pose])
```

5. Phase 3: Mathematical Normalization & Augmentation

5.1 Coordinate Normalization Logic

To remove variations caused by relative distance to the camera lens:

1. **Origin Re-Centering:** Shift all hand coordinates relative to the wrist point:

   $$x' = x - x_{\text{wrist}}, \quad y' = y - y_{\text{wrist}}, \quad z' = z - z_{\text{wrist}}$$

2. **Scale Invariance Scaling:** Normalize all vectors by dividing coordinates by D_{max}, where D_{max} is the distance between the wrist and the middle finger MCP joint.

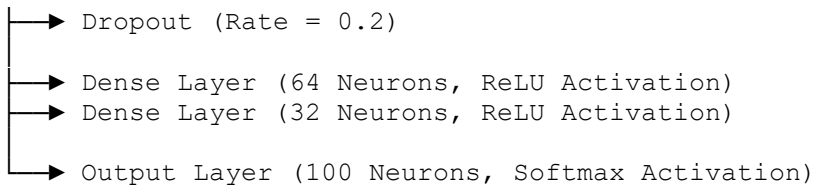
5.2 Data Augmentation Techniques

- **Spatial Rotation:** Apply random 2D plane rotations between -12° and $+12^\circ$.
- **Gaussian Noise:** Inject mild noise ($\sigma = 0.003$) to 15% of frames to simulate minor camera sensor instability.
- **Temporal Interpolation:** Downsample/upsample sequence length to account for fast versus slow hand movements.

6. Phase 4: Model Architecture & Training Setup

6.1 Neural Network Topology

```
Input Layer: Shape (30 Frames, 159 Features)
   |
   |---> LSTM Layer 1 (64 Units, Return Sequences = True)
   |---> Batch Normalization + Dropout (Rate = 0.2)
   |
   |---> LSTM Layer 2 (128 Units, Return Sequences = True)
   |---> Batch Normalization + Dropout (Rate = 0.3)
   |
   |---> LSTM Layer 3 (64 Units, Return Sequences = False)
```



6.2 Training Hyperparameters

- **Loss Function:** Categorical Cross-Entropy
- **Optimizer:** Adam ($\text{Learning Rate} = 0.001$, decaying dynamically on plateau)
- **Batch Size:** 32
- **Epoch Limit:** 200 (with Early Stopping at patience $= 15$)
- **Data Split:** 70% Training, 15% Validation, 15% Testing

7. Phase 5: Real-Time Inference & Output Smoothing

1. **FIFO Frame Buffer:** A rolling buffer maintains the last 30 frames (1 second of video at 30 FPS).
2. **Confidence Thresholding:** Model outputs with less than 85% probability are discarded.
3. **Debounce Logic:** A label must hold continuous dominance across at least **5 consecutive frame evaluations** before being rendered as text to the screen.
4. **Cooldown Delay:** A 1.5-second lock prevents duplicate text outputs while the patient lowers their hands.

8. Phase 6: Edge Optimization & Privacy Safeguards

8.1 Model Quantization

To ensure smooth execution on low-cost tablets without internet access:

- Models are converted to **TensorFlow Lite (`.tflite`)** with **INT8 Quantization**.
- Binary file size drops from $\sim 18\text{ MB}$ to $\sim 2.8\text{ MB}$, yielding a $3.5\times$ speed up on standard mobile processors.

8.2 Privacy by Design Protocol

- **Zero Video Storage:** Video frames are held in memory solely for MediaPipe extraction and immediately overwritten. No audio or video data is stored or uploaded.